

06 — Mission Crane 3000 kW (budget devoured, plant reshuffles)

Test 01's vessel + **Mission Max** = **3000** (battery 1260, Transit).

Battery Contribution

Row	H	I	J	L
Mission (D=0.5)	3 000	$\min(1260, 3000) = \mathbf{1\ 260}$ (ALL of it)	630	2 370
Propulsion	573.15	$\min(0, \dots) = \mathbf{0}$	0	573.15
Hotel	76	0	0	76

Tiles: **PS** = **630** · **SR** = **3 019.2**. The crane, third in priority, drains the budget; everyone below starves.

Power Demands — the reshuffle

Hotel' = $3\ 800 + 2\ 370 + 76 = \mathbf{6\ 246}$ — MORE than one SG (3 250) can carry. Options: 1 ME+SG+AEs (SG 3 250 + AE 2 996 — expensive aux running), or **2 ME + both SGs (6 500) + 0 AE**.

The optimizer picks the second: ME = $12\ 036.15 + 6\ 246 = \mathbf{18\ 282}$ @ 38.1 %, **AE = 0**, SG at 96.1 % ($6\ 246/6\ 500$). Energy ME $18\ 282.15 \times 5\ 000 = \mathbf{91\ 410\ 750}$.

Baseline & IL1

Sorted: **3.1095 (2/on/0 — now FIRST!)** / 3.1292 (1/1) / **3.1780 (1/2 — baseline)** / 3.2102 / 3.2210. The combo that was the WORST row in test 01 is now the best — the uncovered reserve changed which configuration wins, not just how much fuel it burns.

- Baseline = $3.1780 \times 5\ 000 = \mathbf{15\ 890\ t/yr}$ (AE 2 996/8 000 = 37.5 %)
- IL1 = $3.1095 \times 5\ 000 = \mathbf{15\ 547.3}$ · **Savings 342.7 t = \$267 335** — 16× test 01's, because the candidate rows are now genuinely different machines, not ±1 idle AE.

Battery Benefit & IL3

Benefit = covered 630 monetized: **631.3 t/yr = \$492 407** — the ceiling for this battery (630 = the whole 1260 × 50 %). IL3 detail: the mission covered band (630, hotel side) fully absorbs the ±500 DRC variation → Variation shows **0**, batteryShaved **500**, L3 adds **0** (anti-double-counting, rule Q4/D4).

Takeaway: the cascade runs BEFORE combination enumeration for a reason — uncovered reserve can flip the optimal plant configuration entirely.